Activity

Visit a nearby health center to observe the analysis of urine. Dip strips can be used to test urine for a range of different factors such as pH, glucose, ketones and proteins. Dip sticks used for detecting glucose contain two enzymes namely, glucose oxidase and peroxidase. These two enzymes are immobilized on a small pad at one end of the stick. The pad is immersed in urine. If the urine contains glucose, a brown coloured compound is produced. The resulting colour pad is matched against a colour chart. The colour does not indicate the current blood glucose concentrations.

CASE STUDY

Both the kidneys of Ravi (28 years) were not functioning and he was undergoing dialysis. He was admitted to a hospital with renal failure. His mother Suganthi (47 years) was willing to donate one of her kidneys to her son after she was given counseling. Their blood groups were matching and later approval was obtained from transplant committee and technical committee. Operation was performed for 5 hrs. He was administered with immunosuppressive drugs and anti inflammatory drugs. He recovered from the operation and returned home.

1. Name the disease Ravi was suffering from.
2. What relation is the donor of the kidney
3. Name the type of matching done to perform the transplant.
4. Why approval has to be got from transplant committee and technical committee?
5. What do you think about Suganthi donating her kidney?

Evaluation



1. Arrange the following structures in the order that a drop of water entering the nephron would encounter them.
   a. Afferent arteriole
   b. Bowman's capsule
   c. Collecting duct
   d. Distal tubule
   e. Glomerulus
   f. Loop of Henle
   g. Proximal tubule
   h. Renal pelvis
2. Name the three filtration barriers that solutes must come across as they move from plasma to the lumen of Bowman's capsule. What components of the blood are usually excluded by these layers?
3. What forces promote glomerular filtration? What forces opposes them? What is meant by net filtration pressure?
4. Identify the following structures and explain their significance in renal physiology?
   a. Juxtaglomerular apparatus
   b. Podocytes
   c. Sphincters in the bladder
5. In which segment of the nephron most of the re-absorption of substances takes place?

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